UEC731: GRAPHICS AND VISUAL COMPUTING

L	T	P	Cr
3	0	0	3.0

Course Objective: Detailed study of computer graphics, 2 D and 3 D transformations, representations and visualization.

Syllabus:

Fundamentals of Computer Graphics: Applications of computer Graphics in various, Video Display Devices, Random scan displays, raster scan displays, DVST, Flat Panel displays, I/O Devices.

Graphics Primitives: Algorithms for drawing Line, circle, ellipse, arcs & sectors, Boundary Fill & Flood Fill algorithm, Color Tables

Transformations & Projections: 2D & 3D Scaling, Translation, rotation, shearing & reflection, Composite transformation, Window to Viewport transformation, Orthographic and Perspective Projections.

Clipping: CohenSutherland, Liang Barsky, Nicholl-Lee-Nicholl Line clipping algorithms, Sutherland Hodgeman, Weiler Atherton Polygon clipping algorithm.

Three Dimensional Object Representations: 3D Modeling transformations, Parallel & Perspective projection, Clipping in 3D. Curved lines & Surfaces, Spline representations, Spline specifications, Bezier Curves & surfaces, B-spline curves & surfaces, Rational splines, Displaying Spline curves & surfaces.

Basic Rendering: Rendering in nature, Polygonal representation, Affine and coordinate system transformations, Visibility and occlusion, depth buffering, Painter's algorithm, ray tracing, forward and backward rendering equations, Phong Shading per pixel per vertex Shading.

Visualization: Visualization of 2D/3D scalar fields: color mapping, iso surfaces. Direct volume data rendering: ray-casting, transfer functions, segmentation. Visualization of: Vector fields and flow data, Time-varying data, High-dimensional data: dimension reduction, parallel coordinates, Non-spatial data: multi-variate, tree/graph structured, text Perceptual and cognitive foundations, Evaluation of visualization methods, Applications of visualization, Basic Animation Techniques like traditional, keyframing.

Familiarization with standards: IEEE 3110-2025 (Computer Vision Algorithms): IEEE 3079.3.1-2025 (Digital Humans).

Course Learning Outcomes (CLO): The student will be able to:

1. Comprehend the concepts related to basics of computer graphics and visualization.
2. Demonstrate various graphics primitives and 2-D, 3-D geometric transformations
3. Understand various clipping techniques
4. Comprehend the concepts related to three dimensional object representations.
5. Implement various hidden surface removal techniques.

Text Books:

1. Donald D Hearn, M. Pauline Baker, Computer Graphics C version, Pearson Education

1. Dave Shreiner, Mason Woo, Jackie Neider, Tom Davis, OpenGL Programming Guide: The Official Guide to Learning OpenGL, (2013).

Reference Books:

1. James D. Foley, Andries van Dam, Steven K. Feiner, John F. Hughes, Computer Graphics: Principles & Practice in C, Addison Wesley Longman.

2. Zhigang Xiang, Roy A Plastock, Computer Graphics, Schaums Outline, TMH